



SPI-SAR-ADC Controller: Digital Design, Verification and Implementation

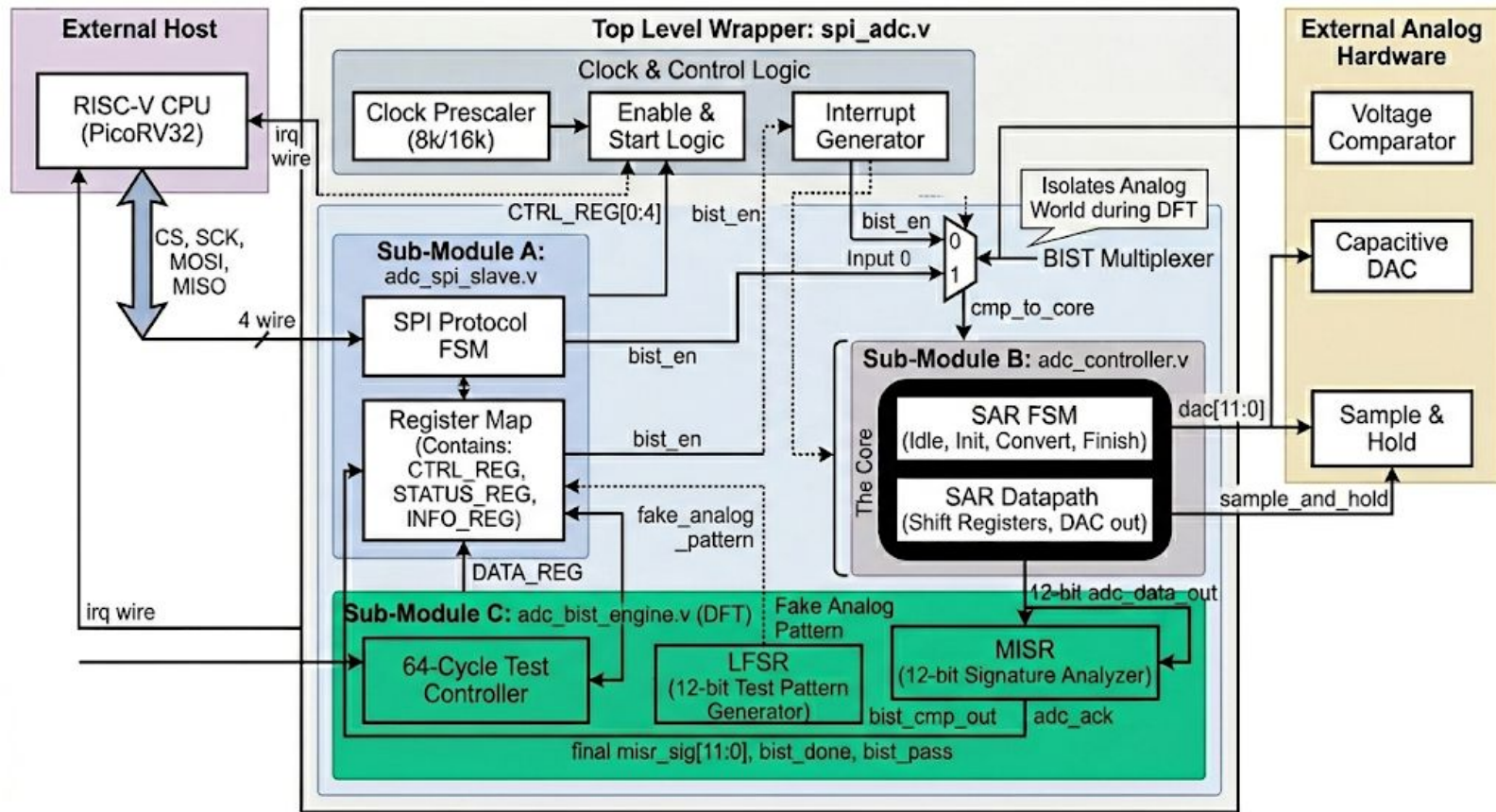
by Saumya Raj Singh



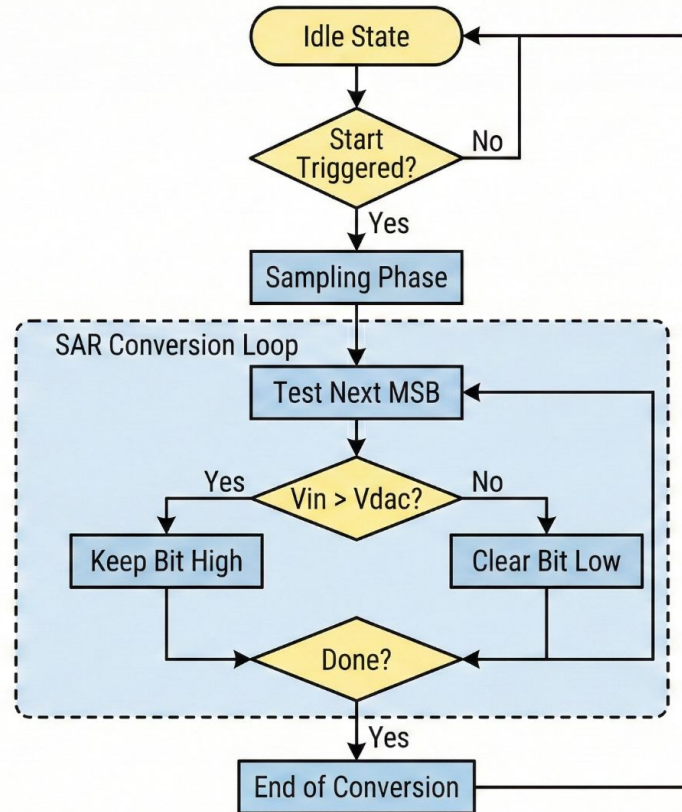
Design and Architecture

System-Level Overview & Modular Breakdown

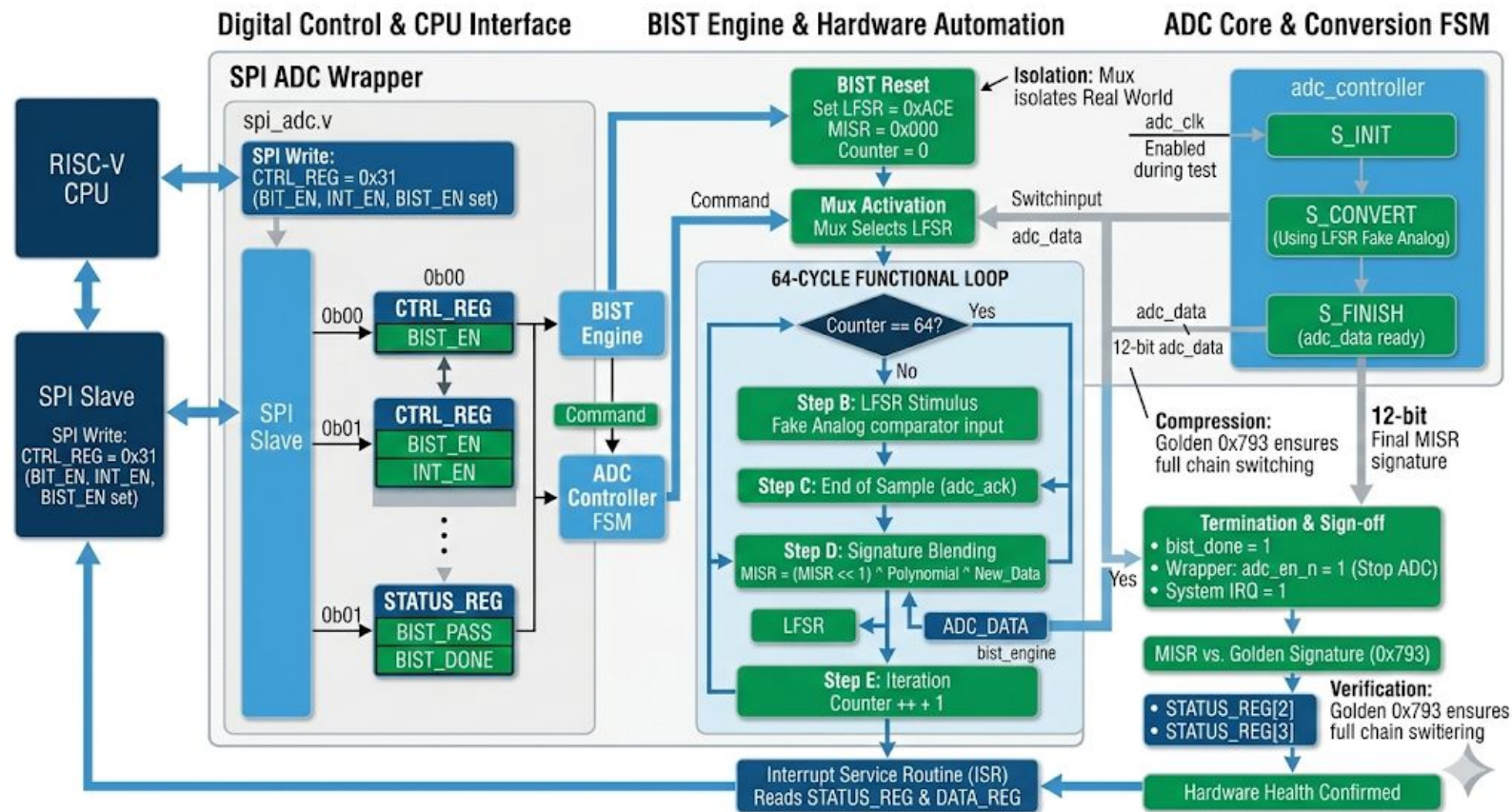
The architecture is partitioned into a top-level spi_adc wrapper and a dedicated ADC controller core to separate host communication from signal acquisition. The SPI module operates as a slave using a standard 4-wire bus (CS, SCK, MOSI, MISO) to manage the internal register map, which handles control, status, and data storage. The ADC controller utilizes a main Finite State Machine (FSM) to drive the Successive Approximation Register (SAR) datapath, coordinating the external DAC and comparator to resolve the 12-bit digital output. To optimize energy consumption, the clock generation block includes a prescaler and clock gating logic, ensuring the high-speed ADC clock is active only during conversion phases.



SYSTEM BLOCK DIAGRAM



ADC CONTROLLER STATE DIAGRAM : SAR CONVERSION ALGORITHM



BUILT-IN SELF TEST

SPI COMMANDS AND REGISTERS DATASHEET

TABLE 1: REGISTER SUMMARY (0b00 to 0b11)

Address	Register Name	R/W	Description
0b00	CTRL_REG	R/W	Main control register (Execution & Test Modes).
0b01	STATUS_REG	R	Combined status flags, busy states, and BIST diagnostics
0b10	DATA_REG	R	Holds the 12-bit conversion result or BIST Signature
0b11	INFO_REG	R	Stores hardcoded Chip ID / Version information.

SPI FRAME STRUCTURE (16-BIT)

Bits [15:14]	Bits [13:12]	Bits [11:0]
CMD (2 bits)	ADDR (2 bits)	PAYLOAD (12 bits)
Commands: Read (00, Write (01, Set (10, Clear 11)	Selects target register 00 to 11	Write data, ignored during Read commands

TABLE 2.1: 0b00: CTRL_REG (ADC Control Register)

Bit(s)	Name	Description
Bits [11:7]	Reserved	Read as 0
Bit 6	CLK_SEL	0 = 8 kHz, 1 = 16 kHz
Bit 5	BIST_EN	1 = Start Built-in Self-Test (LFSR/MISR Mode)
Bit 4	INT_EN	1 = Interrupt Enable, 0 = Disable
Bit 3	VREF_SEL	1 = External V_{ref} , 0 = Internal V_{ref}
Bit 2	AUTO	1 = Auto Mode, 0 = Manual Mode
Bit 1	START	1 = Start Conversion (Self-clearing)
Bit 0	ADC_EN	1 = ADC Enable, 0 = Global Shutdown

TABLE 2.1: 0b00: CTRL_REG (ADC Control Register)

Bit(s)	Name	Description
Bits [11:7]	Reserved	Read as 0
Bit 6	CLK_SEL	0 = 8 kHz, 1 = 16 kHz
Bit 5	BIST_EN	1 = Start Built-In Self-Test (LFSR/MISR Mode)
Bit 4	INT_EN	1 = Interrupt Enable, 0 = Disable

TABLE 2.2: 0b01: STATUS_REG (ADC Status Register)

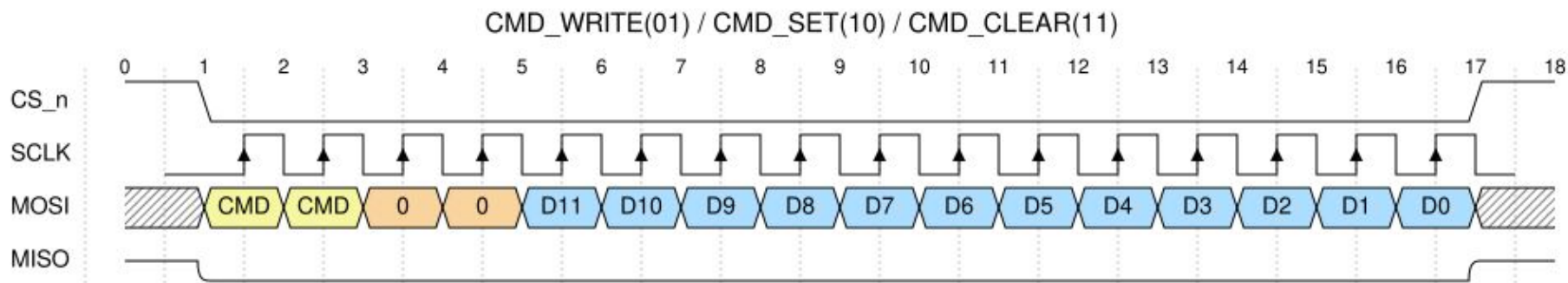
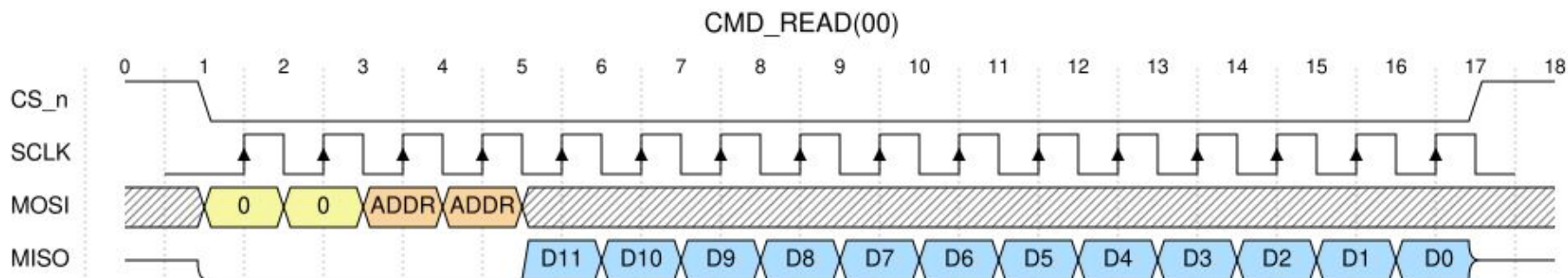
Bit(s)	Name	Description
Bits [11:4]	Reserved	Read as 0
Bit 3	BIST_PASS	1 = MISR Signature verified perfectly.
Bit 2	BIST_DONE	1 = 64-cycle Automated BIST completed.
Bit 1	BUSY	1 = Conversion or BIST in progress.
Bit 0	EOC	End of Conversion/Test. Clears on read.

TABLE 2.3: 0b10: DATA_REG (ADC Data Register)

Bit(s)	Name	Description
Bits [11:0]	ADC_DATA	The 12-bit conversion result OR the final 12-bit BIST Signature.

TABLE 2.4: 0b11: INFO_REG (Info Register)

Bit(s)	Name	Description
Bits [11:0]	CHIP_ID	Hardcoded Chip ID (e.g., 0x414 for 'A')



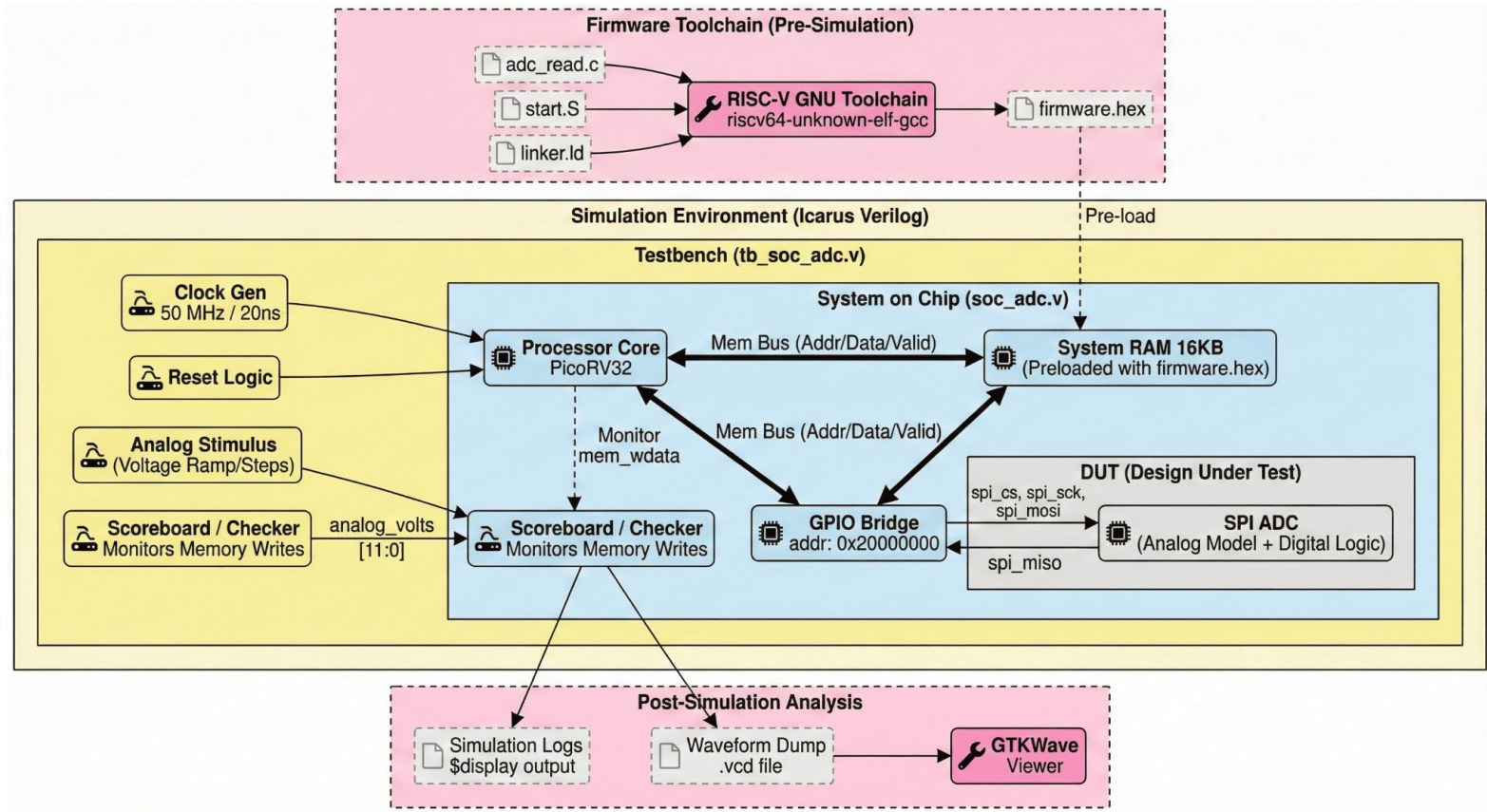
SPI TIMING DIAGRAM



Functional Verification

SoC-Level Validation via
Hardware-Firmware
Co-Simulation

Verification is conducted within a complete SoC environment where a PicoRV32 processor core executes C-based firmware (`adc_read.c`) to drive the Design Under Test (DUT). The testbench architecture integrates a sophisticated scoreboard and checker that monitor memory writes via the system bus to verify that digital outputs match the analog input stimuli. Extensive testing covers multiple scenarios, including register Read/Write checks, interrupt generation, and auto-sampling modes, ensuring the logic correctly handles both single-shot and continuous conversion requests under varied timing conditions.



TESTBENCH ARCHITECTURE

===== STARTING EXHAUSTIVE TESTS =====

TEST 1: Defaults

[PASS]

Reset CTRL: 0x000

[PASS]

Info Reg: 0x00a

TEST 2: Register R/W

[PASS]

Readback: 0x5a5

TEST 3: VREF Pin

[PASS] VREF High

TEST 4: Single 8ksps (Input 0xA52)

[PASS] 8ksps Timing OK (120010 ns)

[PASS]

Result: 0xa52

TEST 5: Data Corners

[PASS]

Min Input: 0x000

[PASS]

Max Input: 0xfff

TEST 6: Self-Clearing Start

[PASS] Bit Cleared

TEST 7: Busy Flag

[PASS] Busy High

TEST 8: Interrupt Masking

[PASS] IRQ Low

[PASS] EOC Set

TEST 9: 16ksps Clock

[PASS] 16ksps Timing OK (59520 ns)

[PASS]

16ksps Data: 0x888

TEST 10: Revert 8ksps

[PASS] 8ksps Timing OK (120000 ns)

[PASS]

8ksps Data: 0x111

TEST 11: Auto Mode (NO IRQ)

[PASS]

Auto 1: 0x200

[PASS]

Auto 2: 0x999

TEST 12: Built-In Self-Test (BIST)

[INFO] BIST completed in 7996360 ns

[PASS] BIST Status flags correct (Done & Pass)

[PASS]

BIST MISR Signature: 0x793

TEST 13: Global Disable

[PASS] Inactive

===== ALL TESTS PASSED =====

BLOCK-LEVEL VERIFICATION RESULTS

BIST ENGINE: CYCLE-ACCURATE MISR SIGNATURE GENERATION										
Conv	Cycle	LFSR Seed	ADC Out	MISR Blending Math (Shift ^ Poly ^ Data)				New MISR		
1	14	0x59C	0xB39	0x000	^	0x000	^	0xB39		0xB39
2	28	0xEAD	0xD5A	0x672	^	0x829	^	0xD5A		0x301
3	42	0xB6F	0x6DF	0x602	^	0x000	^	0x6DF		0x0DD
4	56	0x906	0x20D	0x1BA	^	0x000	^	0x20D		0x3B7
5	70	0x345	0x68A	0x76E	^	0x000	^	0x68A		0x1E4
6	84	0x6CA	0xD95	0x3C8	^	0x000	^	0xD95		0xE5D
7	98	0x293	0x526	0xCBA	^	0x829	^	0x526		0x1B5
8	112	0x6A7	0xD4F	0x36A	^	0x000	^	0xD4F		0xE25
9	126	0xB20	0x641	0xC4A	^	0x829	^	0x641		0x222
10	140	0x910	0x220	0x444	^	0x000	^	0x220		0x664
11	154	0x81F	0x03F	0xCC8	^	0x000	^	0x03F		0xCF7
12	168	0x0D1	0x1A3	0x9EE	^	0x829	^	0x1A3		0x064
13	182	0x03B	0x076	0x0C8	^	0x000	^	0x076		0x0BE
14	196	0xDEB	0xBDF	0x17C	^	0x000	^	0xBDF		0xAA3
15	210	0xA2D	0x45A	0x546	^	0x829	^	0x45A		0x935
16	224	0x7F8	0xFF0	0x26A	^	0x829	^	0xFF0		0x5B3
17	238	0xDD8	0xBB1	0xB66	^	0x000	^	0xBB1		0x0D7
18	252	0x316	0x62D	0x1AE	^	0x000	^	0x62D		0x783
19	266	0x338	0x671	0xF06	^	0x000	^	0x671		0x977
20	280	0xE49	0xC93	0x2EE	^	0x829	^	0xC93		0x654
21	294	0x449	0x892	0xCA8	^	0x000	^	0x892		0x43A
22	308	0x504	0xA08	0x874	^	0x000	^	0xA08		0x27C
23	322	0x4EA	0x9D5	0x4F8	^	0x000	^	0x9D5		0xD2D
24	336	0x2AC	0x558	0xA5A	^	0x829	^	0x558		0x72B
25	350	0x704	0xE08	0xE56	^	0x000	^	0xE08		0x05E
26	364	0xB56	0x6AD	0x0BC	^	0x000	^	0x6AD		0x611
27	378	0x2CF	0x59E	0xC22	^	0x000	^	0x59E		0x9BC
28	392	0xC45	0x88B	0x378	^	0x829	^	0x88B		0x3DA
29	406	0xF21	0xE42	0x7B4	^	0x000	^	0xE42		0x9F6
30	420	0xB8A	0x715	0x3EC	^	0x829	^	0x715		0xCD0
31	434	0xBC2	0x784	0x9A0	^	0x829	^	0x784		0x60D
32	448	0xC5C	0x8B9	0xC1A	^	0x000	^	0x8B9		0x4A3
33	462	0xB6B	0x6D7	0x946	^	0x000	^	0x6D7		0xF91

33	462	0xB6B	0x6D7	0x946	^	0x000	^	0x6D7		0xF91
34	476	0x54A	0xA95	0xF22	^	0x829	^	0xA95		0xD9E
35	490	0x91E	0x23C	0xB3C	^	0x829	^	0x23C		0x129
36	504	0xAED	0x5DB	0x252	^	0x000	^	0x5DB		0x789
37	518	0xB11	0x623	0xF12	^	0x000	^	0x623		0x931
38	532	0xA41	0x483	0x262	^	0x829	^	0x483		0xEC8
39	546	0x3A9	0x753	0xD90	^	0x829	^	0x753		0x2EA
40	560	0x174	0x2E8	0x5D4	^	0x000	^	0x2E8		0x73C
41	574	0xA27	0x44E	0xE78	^	0x000	^	0x44E		0xA36
42	588	0x946	0x28C	0x46C	^	0x829	^	0x28C		0xEC9
43	602	0xC43	0x887	0xD92	^	0x829	^	0x887		0xD3C
44	616	0x54B	0xA96	0xA78	^	0x829	^	0xA96		0x8C7
45	630	0x4FC	0x9F8	0x18E	^	0x829	^	0x9F8		0x05F
46	644	0x9F6	0x3ED	0x0BE	^	0x000	^	0x3ED		0x353
47	658	0x11F	0x23E	0x6A6	^	0x000	^	0x23E		0x498
48	672	0x9FE	0x3FD	0x930	^	0x000	^	0x3FD		0xADC
49	686	0x987	0x30E	0x59A	^	0x829	^	0x30E		0xEBD
50	700	0xD48	0xA90	0xD7A	^	0x829	^	0xA90		0xFC3
51	714	0x1C9	0x392	0xF86	^	0x829	^	0x392		0x43D
52	728	0xE4D	0xC9B	0x87A	^	0x000	^	0xC9B		0x4E1
53	742	0x805	0x00A	0x9C2	^	0x000	^	0x00A		0x9C8
54	756	0xF5F	0xEBE	0x390	^	0x829	^	0xEBE		0x507
55	770	0x8CD	0x19B	0xA0E	^	0x000	^	0x19B		0xB95
56	784	0xB2E	0x65D	0x72A	^	0x829	^	0x65D		0x95E
57	798	0xBE2	0x7C4	0x2BC	^	0x829	^	0x7C4		0xD51
58	812	0x3DF	0x7BF	0xAA2	^	0x829	^	0x7BF		0x534
59	826	0xAAB	0x556	0xA68	^	0x000	^	0x556		0xF3E
60	840	0xE7D	0xCFA	0xE7C	^	0x829	^	0xCFA		0xAAB
61	854	0x6B6	0xD6D	0x55E	^	0x829	^	0xD6D		0x01A
62	868	0x7F2	0xFE4	0x034	^	0x000	^	0xFE4		0xFD0
63	882	0x366	0x6CD	0xFA0	^	0x829	^	0x6CD		0x144
64	896	0x28D	0x51B	0x288	^	0x000	^	0x51B		0x793

--- BIST SIMULATION SUMMARY ---

Total Clock Cycles : 897

Total Conversions Run : 64

Final Golden MISR : 0x793

THEORETICAL PYTHON SIMULATION FOR BIST

```

-----
1. Building Firmware (adc_read.c)...
-----
🔧 Compiling for RISC-V (RV32I)
Script Root: /home/kyouma-hououin/Documents/VLSI/designs/spi_sar_adc/riscv_toolchain
Source:      riscv_toolchain/source/adc_read.c
Helpers:     start.S, linker.ld (in Script Root)
Output:      /home/kyouma-hououin/Documents/VLSI/designs/spi_sar_adc/riscv_toolchain/firmware/firmware.hex
-----
/usr/lib/gcc/riscv64-unknown-elf/13.2.0/../../../../riscv64-unknown-elf/bin/ld: warning: /home/kyouma-hououin/Documents/VLSI/designs/spi_sar_adc/riscv_toolchain/firmware/firmware.elf has a LOAD segment with R/WX permissions
✅ Success! Hex file generated at:
/home/kyouma-hououin/Documents/VLSI/designs/spi_sar_adc/riscv_toolchain/firmware/firmware.hex
-----
2. Compiling Verilog Hardware...
-----
3. Running Simulation...
-----
VCD info: dumpfile tb_soc_adc.vcd opened for output.

--- ADC SOC SIMULATION (COMPREHENSIVE) START ---

[RAM] Firmware wrote Result:      2566
[Time 831350000] CPU wrote      2566 (0x00000a06) to 0x00000400
✅ SINGLE SHOT (16kHz) PASSED! (Got 2566) (Expected 2567)

-> Changing Input to 1543 for Auto Mode (8kHz)...
[Time 2094210000] CPU wrote      1542 (0x00000606) to 0x00000404
✅ AUTO SAMPLE 1 (8kHz) PASSED! (Got 1542) (Expected 1543)

-> Changing Input to 2654...
[Time 2948470000] CPU wrote      2653 (0x00000a5d) to 0x00000408
✅ AUTO SAMPLE 2 (8kHz) PASSED! (Got 2653) (Expected 2654)

-> Changing Input to 3129...
[Time 3800210000] CPU wrote      3128 (0x00000c38) to 0x0000040c
✅ AUTO SAMPLE 3 (8kHz) PASSED! (Got 3128) (Expected 3129)

-> Starting Abort Test (Disabling ADC)...
[Time 4642110000] CPU wrote      43783 (0x0000ab07) to 0x00000410
✅ ABORT TEST PASSED! Busy flag cleared.

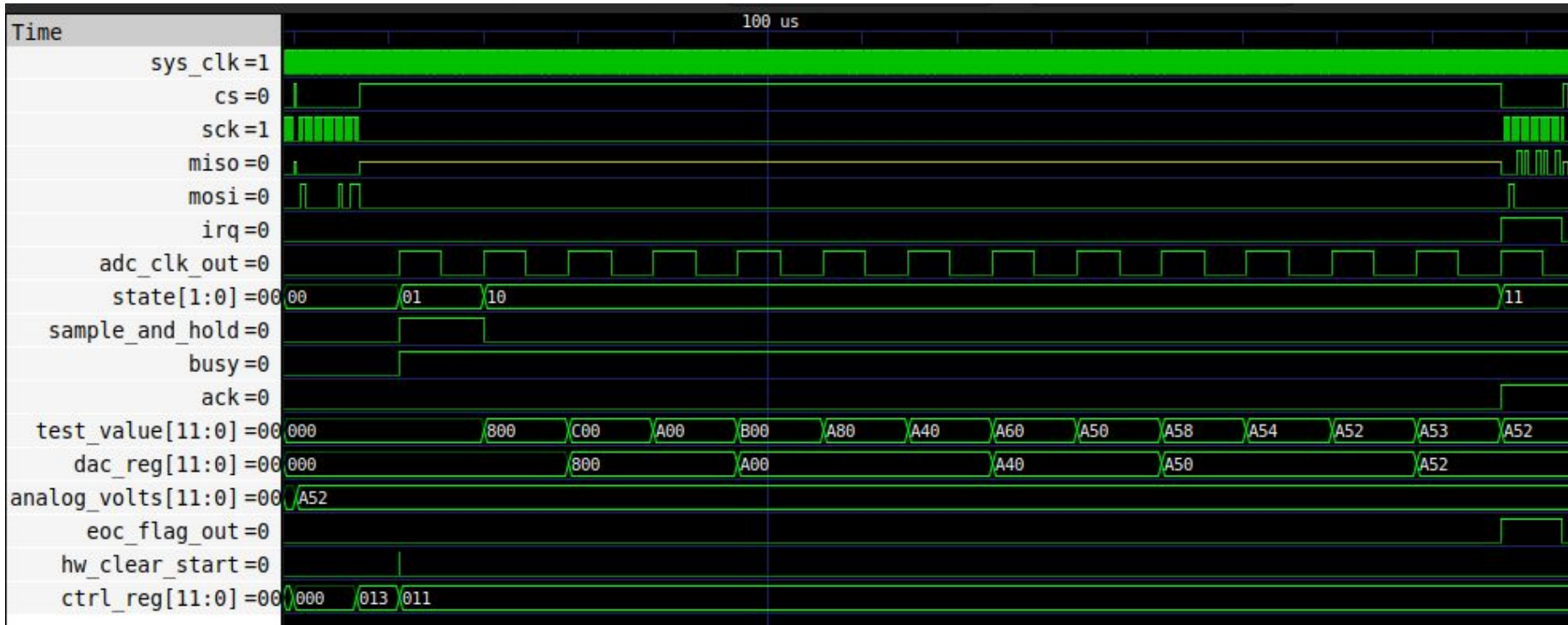
🎉 ALL TESTS PASSED (Single, Auto, Clock, Abort)

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SOC-LEVEL VERIFICATION RESULTS



ADC SINGLE SHOT CONVERSION WAVEFORM: START PHASE



ADC SINGLE SHOT CONVERSION WAVEFORM: CONVERSION PHASE



ADC SINGLE SHOT CONVERSION WAVEFORM: END PHASE

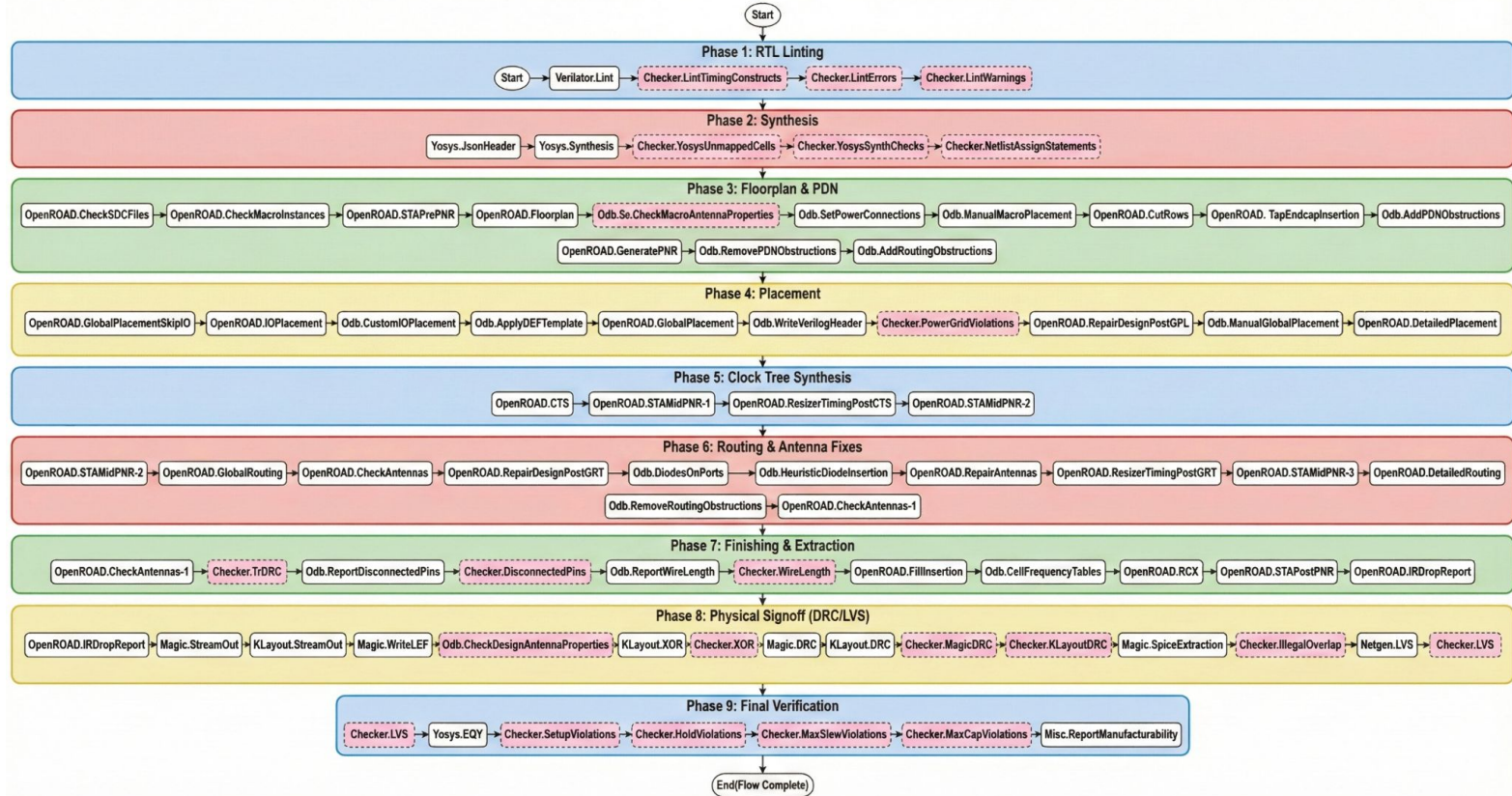


Physical Implementation

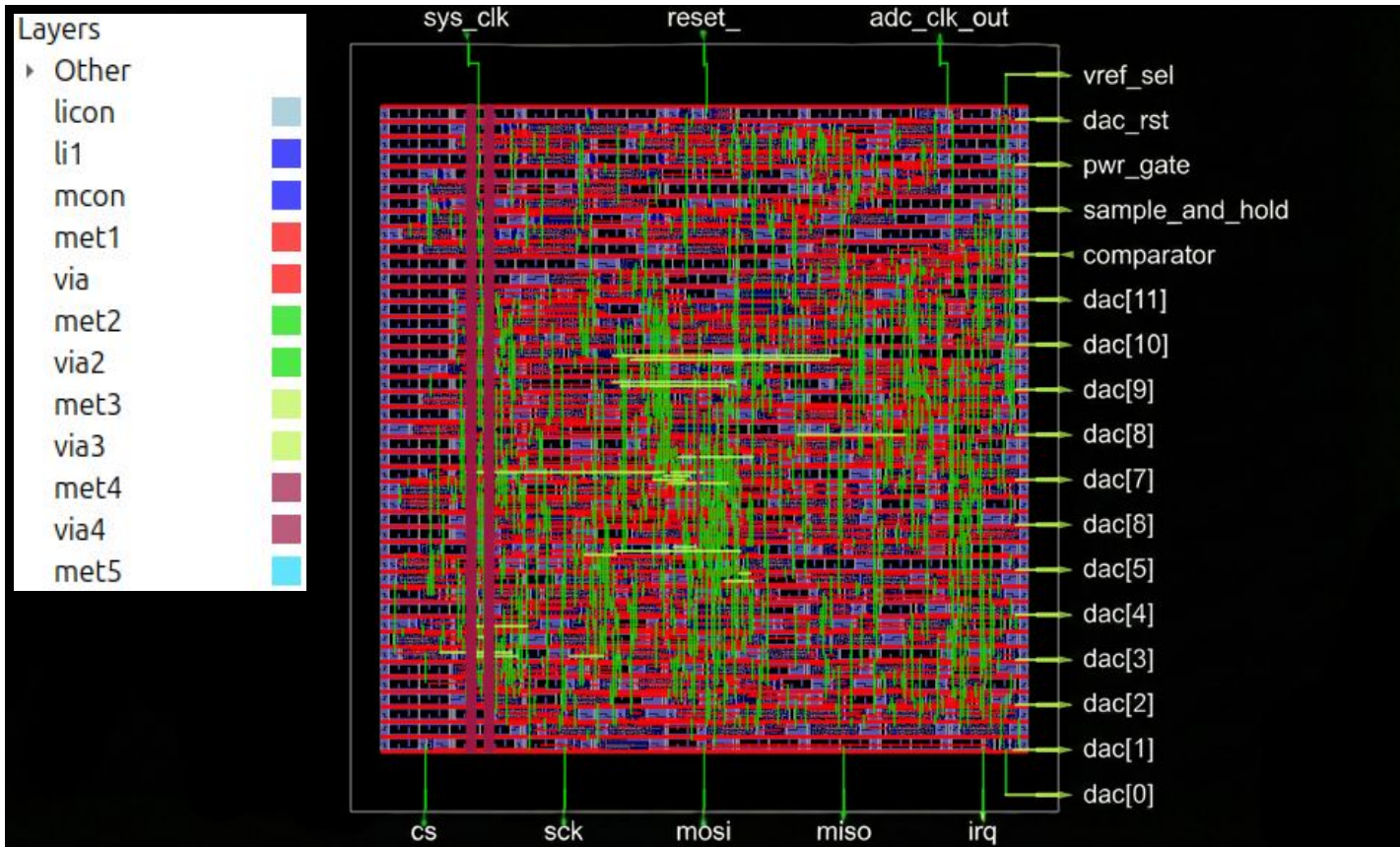
Automated RTL-to-GDSII Flow &
Physical Signoff

The physical design is implemented using the LibreLane "Classic" flow, which automates the progression from RTL linting and synthesis (Yosys) to floorplanning, placement, and detailed routing (OpenROAD). The layout undergoes Multi-Corner Static Timing Analysis (STA) to verify setup and hold compliance across different process variations, alongside physical verification (DRC/LVS) using Magic and Netgen to ensure manufacturability. A final comparative Power, Performance, and Area (PPA) analysis confirmed the optimal configuration, achieving a total power consumption of 108.9 μ W at 10Mhz clk by utilizing high-density cells and active clock gating.

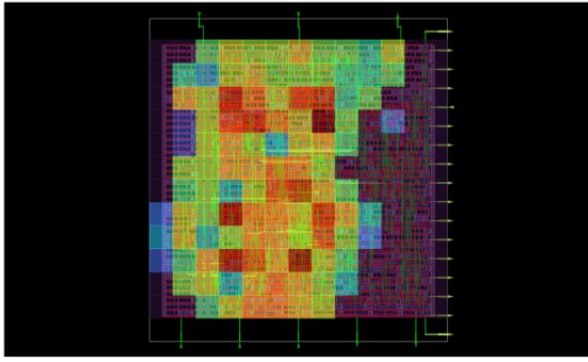
LibreLane Classic Flow



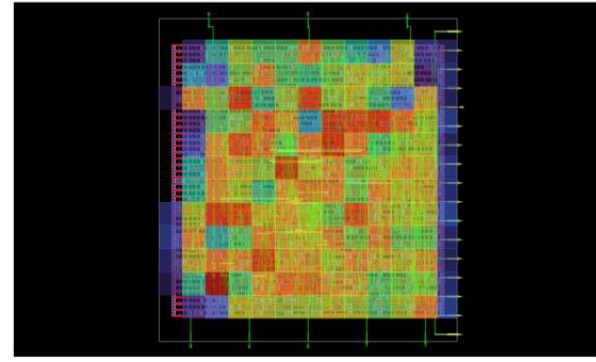
LIBRELANE "CLASSIC" SYNTHESIS FLOW



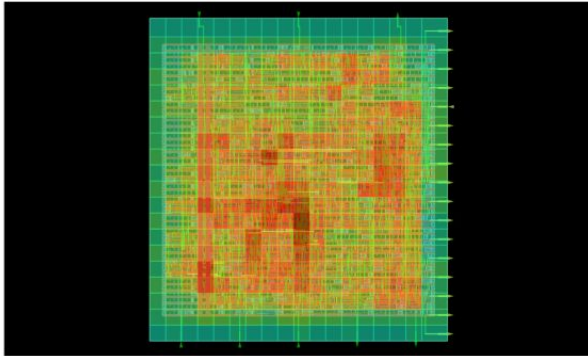
OPENROAD DATABASE FORMAT (OBD) VIEW



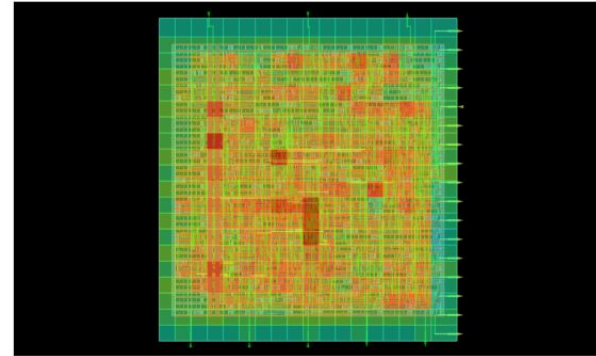
(a) Power Density Map



(b) Placement Density Map

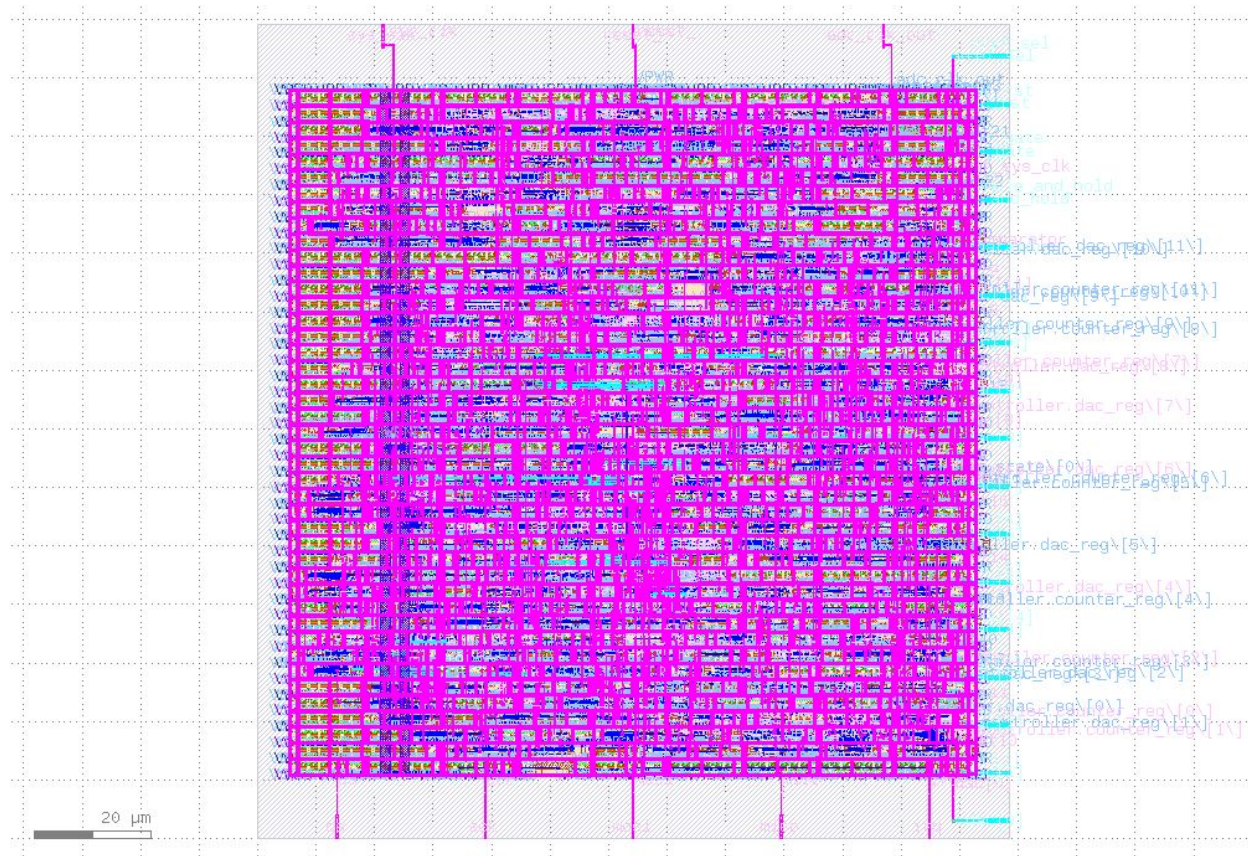


(c) Estimated Congestion (RUDY)



(d) Routing Congestion

HEAT MAP ANALYSIS



KLAYOUT GRAPHIC DATA SYSTEM (GDS) VIEW

Constraint: CLOCK_PERIOD = 100.0 ns (10 MHz).

Power Component	Value	Percentage
Total Power	108.9 μW	100.0%
Internal Power	79.8 μW	73.3%
Switching Power	29.1 μW	26.7%
Leakage Power	8.28 nW	< 0.01%

- **DRC:** 0 Violations (Magic).
- **LVS:** 0 Mismatches (Netgen).
- **Antenna:** 0 Violations (OpenROAD/Arc).

Constraint: CLOCK_PERIOD = 100.0 ns (10 MHz).

Corner	Hold WNS (ns)	Hold TNS (ns)	Setup WNS (ns)	Setup TNS (ns)
Overall	0.1081	0.0000	58.9116	0.0000
nom_tt_025C_1v80	0.3147	0.0000	59.2983	0.0000
nom_ss_100C_1v60	0.8657	0.0000	58.9178	0.0000
nom_ff_n40C_1v95	0.1105	0.0000	59.4411	0.0000
min_tt_025C_1v80	0.3114	0.0000	59.3048	0.0000
min_ss_100C_1v60	0.8592	0.0000	58.9283	0.0000
min_ff_n40C_1v95	0.1081	0.0000	59.4456	0.0000
max_tt_025C_1v80	0.3181	0.0000	59.2946	0.0000
max_ss_100C_1v60	0.8715	0.0000	58.9116	0.0000
max_ff_n40C_1v95	0.1129	0.0000	59.4383	0.0000

MULTI_CORNER STATIC TIMING ANALYSIS SUMMARY

- 1. **Config A (Baseline):** 45% Utilization, Clock Gating Enabled.
- 2. **Config B (High Density):** 60% Utilization, Clock Gating Enabled.
- 3. **Config C (Low Power Baseline):** 45% Utilization, Clock Gating Disabled.

Metric	Config A (Base)	Config B (High Dens.)	Config C (No Gating)
Instance Count	747	709	747
Cell Area (μm^2)	7124.33	7129.34	7148.11
Total Power (W)	1.089e-4	1.147e-4	1.093e-4
Switching Power (W)	2.908e-5	3.095e-5	3.071e-5
Setup WNS (ns)	59.30	58.91	59.30
Clock Skew (ns)	0.63	0.58	0.63

Project Repository:

https://github.com/SaumyaRaj188/spi_sar_adc

